

OpenCart Graduation Project: Quality Assurance Report

1. Project Overview & Scope

The OpenCart Graduation Project focused on establishing a robust, automated e-commerce platform. This document serves as the final record for all Quality Assurance and Automation efforts, detailing the framework, execution results, and continuous integration pipeline for a reliable product launch.

2. Team & Resources

The success of this project is the direct result of the dedicated work from the following team members:

Role	Members
Team Leader	Nagham Ahmed Mahmoud Saed
Team Member	Ahmed Gamal Hassan Ali Ghoneim
Team Member	Kareem Fareed Abd-Elrahman Fahmy
Team Member	Jana Mohamed El-Fateh Mohamed
Team Member	Alaa Ahmed Tolba Ashmawy
Team Member	Doha Ramadan Salah El-Khedewy

3. Strategic Objectives

Our quality strategy targeted four key areas to ensure comprehensive application hardening. The status reflects the current state of the **OpenCart application based on our testing outcomes**:

Objective	Rationale	Status
Automate Critical User Journeys	Reduce manual regression effort for core flows (Login, Cart, Checkout).	 Complete

Validate API Endpoints	Guarantee backend stability and data integrity across services.	⚠️ Application Needs Improvement
Verify Database Integrity	Prevent data corruption and ensure transactional consistency.	⚠️ Application Needs Improvement
Implement CI/CD Pipeline	Establish continuous feedback loop for code pushes.	✅ Complete

4. Key Deliverables & Results

This project produced four primary artifacts, forming the foundation of our ongoing quality strategy:

4.1. Optimized Automation Framework

- **Design:** Hybrid approach utilizing the Page Object Model (POM) for maximum maintainability.
- **Technology Stack:** Built on **Java, Selenium, and TestNG**.
- **Stability Feature:** Explicit Waits implemented throughout the framework for highly stable execution.
- **Documentation:** [View Framework Overview](#)

4.2. Full Execution Report Summary

The latest execution cycle shows high UI stability but indicates necessary backend work **in the application under test**.

Metric	Value	Comment
Total Tests Run	304	Comprehensive coverage across user flows.
Overall Pass Rate	93.4%	Strong result, exceeding initial benchmark of 90%.
Key Finding	Strong UI stability, but application's backend validation needs improvement.	Focus area for post-release hardening.

- **Documentation:** [View Full Report \(PDF\)](#)

4.3. CI/CD Integrated Pipeline

- **Platform:** GitHub Actions Workflow.
- **Functionality:** Automates the build and test process, running the full suite upon every code push.
- **Benefit:** Ensures immediate verification and continuous quality checks.
- **Documentation:** [View Pipeline Details](#)

4.4. Final Completion Report

The official sign-off on testing activities.

- **Recommendation:** **Conditional Go** for Release.
- **Primary Risk:** Specific risks identified in **application's** database constraints that must be addressed immediately post-launch.
- **Documentation:** [View Sign-Off \(PDF\)](#)

5. Conclusion & Next Steps

The OpenCart project now stands on a scalable and automated quality foundation, successfully *exposing* critical vulnerabilities in the application under test. The framework is ready for expansion, and the CI/CD pipeline guarantees ongoing security. The immediate next step is to prioritize and address the identified backend validation and database constraint gaps **within the OpenCart application** to fully harden the application's core logic.